MemSQL
Self-Guided
Proof of Concept Documentation

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Document Purpose & Scope

This document is intended to assist MemSQL customers in setting up a test MemSQL database cluster. This document is not a replacement for a well-engineered system designed for a specific workload or requirement.

With this document, MemSQL customers will be able to see firsthand the ease and speed in which they can deploy MemSQL. They will also see that configuring MemSQL for High Availability is extremely trivial and easy.

MemSQL AWS/C2S Minimum Instance Recommendations/Requirements:

The following table shows the recommended resources to use when deploying MemSQL in a cloud environment, including several options to meet different application performance requirements. MemSQL will work with most public or private cloud providers, as long as the selected resource has comparable specifications to those listed below. All instances should have high network performance and must be a supported 64-bit Linux distribution. All instances should be geographically deployed in a single region. Instance types that support Enhanced Networking should have it enabled.

Instance Type	Processor Architecture	vCPU	Memory(GB)	Instance Storage	Network
Small	64-bit	8	7	4x420	High
Medium	64-bit	4	15	4x420	High
Large	64-bit	8	30	EBS only	High
Extra Large	64-bit	8	68.4	2x840	High

Deployment Procedures:

MemSQL offers convenient deployment procedures. These procedures are, in order of preference:

1) MemSQL Ops GUI Deployment
2) MemSQL Ops Command Line Deployment
3) AWS CloudFormation

Pre-Deployment Requirements

To deploy MemSQL, either with the GUI or command line, please make sure that you have the following pre-requites completed:

1) The latest copy of MemSQL Ops (file name: memsql-ops-x.x.x.tar.gz)
2) The latest copy of MemSQL Database binary (file name: memsqlbin_amd64.tar.gz)
3) A copy of your Enterprise License Key
4) Valid username & password credentials for the target servers
5) Ports 3306 and 9000 open. Port 3306 is used by the MemSQL database, and port 9000 is used by MemSQL Ops.

Once these items are complete, do the following at the command on the system that will be acting as the MemSQL Ops primary:

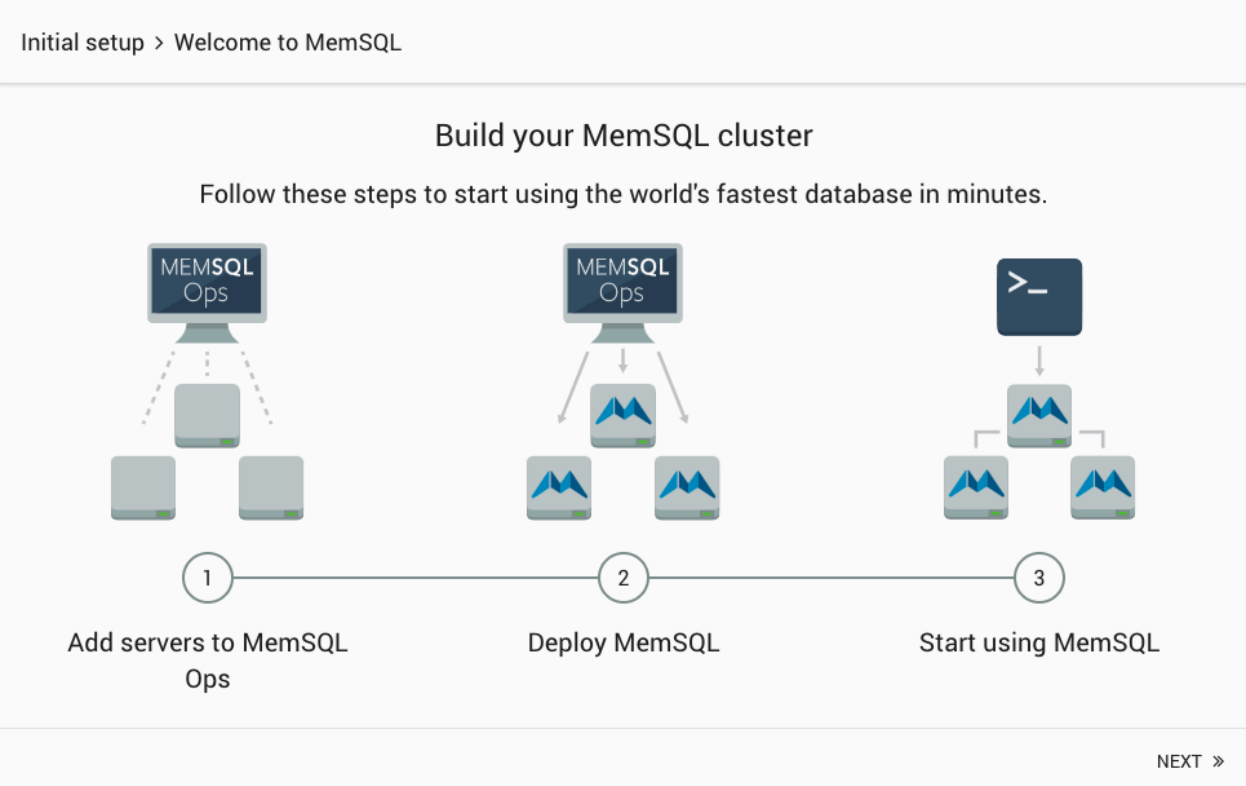
```
tar -xzf memsql-ops-x.x.x.tar.gz
cd memsql-ops-x.x.x
sudo ./install.sh -n
```

MemSQL Ops will complete it's setup and be available on port 9000 of the designated server, i.e. http://<server IP address>:9000.

MemSQL Ops GUI Deployment

To deploy the MemSQL Ops agent, open up MemSQL Ops on the primary server at http://<server IP address>:9000

The Initial Setup screen should appear prompting to add servers to MemSQL Ops. Click the NEXT button.



Next, select the Enterprise Edition of MemSQL

MEMSQL

Initial setup > Welcome to MemSQL

Please select a MemSQL edition.

Enterprise Edition
MemSQL with enterprise features and support.

Community Edition
No signup required. Free forever.

« BACK

Enter the Enterprise Edition license key and click the NEXT button.

Initial setup > Welcome to MemSQL

Already have a license? Enter it here:

LICENSE KEY

Or, start a free Enterprise Edition trial now!

Try Enterprise Edition

Start your 30-day trial and get...

- High Availability
- Cross Data Center Replication
- Granular User Permissions
- 24x7 Support

FIRST NAME

LAST NAME

COMPANY NAME

PHONE NUMBER

EMAIL ADDRESS

Start Trial

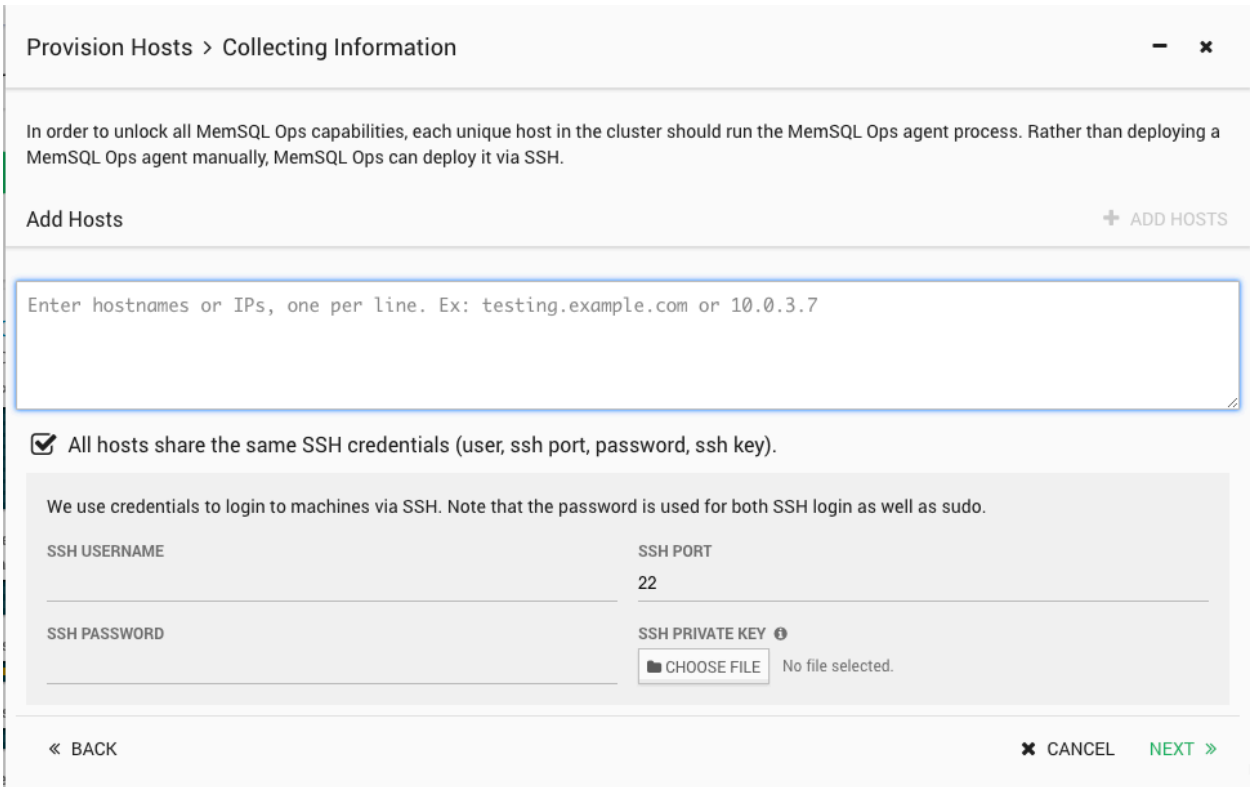
By clicking the button, you agree to the MemSQL terms of service.

« BACK

NEXT »

Once the license key is validated, select to install on Multiple Hosts.

Provision the hosts by entering the IP addresses of ALL hosts for the MemSQL cluster along with SSH Username & Password (or private key).



On the Confirmation Page, select all the hosts and click the PROVISION HOSTS button.

Provision Hosts > Collecting Information

In order to unlock all MemSQL Ops capabilities, each unique host in the cluster should run the MemSQL Ops agent process. Rather than deploying a MemSQL Ops agent manually, MemSQL Ops can deploy it via SSH.

Confirm 9 Target Hosts

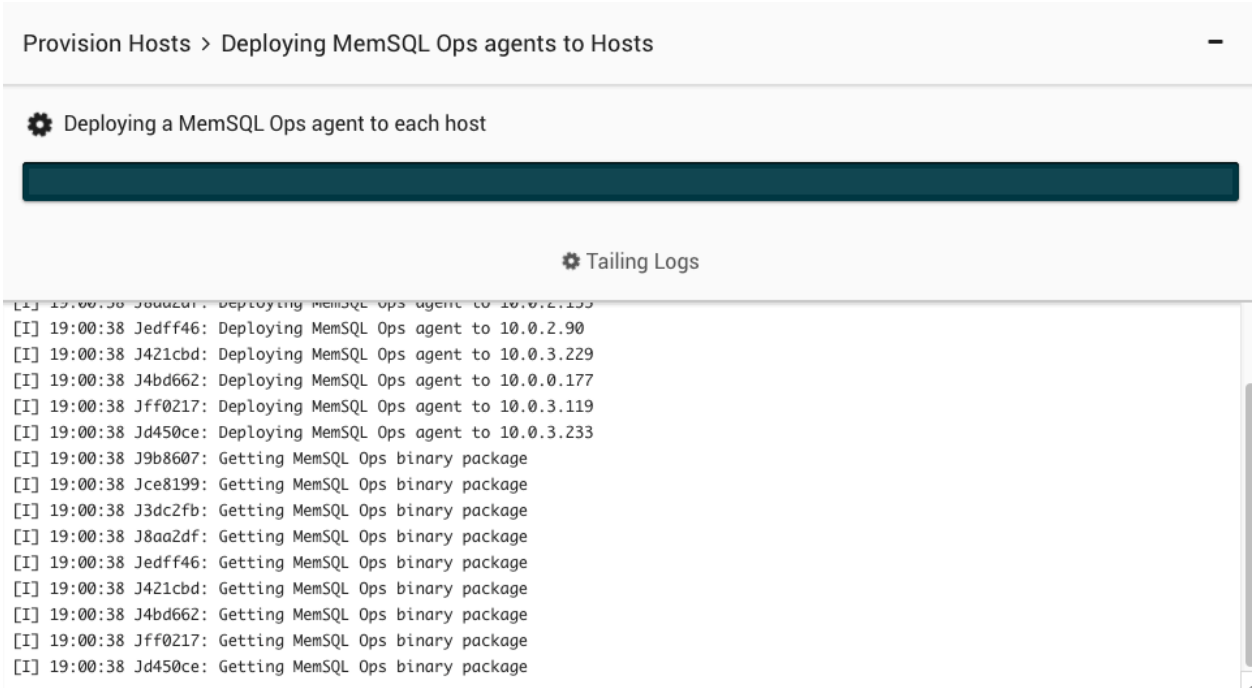
ADD HOSTS

Host	User	SSH Port	Agent Port	Error?
10.0.0.128	admin	22	9000	
10.0.0.177	admin	22	9000	
10.0.0.65	admin	22	9000	
10.0.1.211	admin	22	9000	
10.0.2.155	admin	22	9000	
10.0.2.90	admin	22	9000	
10.0.3.119	admin	22	9000	
10.0.3.229	admin	22	9000	
10.0.3.233	admin	22	9000	

9 Selected Records REMOVE SELECTED EDIT SELECTED

« BACK CANCEL PROVISION HOSTS »

Next MemSQL Ops will check for any errors and begin deploying MemSQL Ops to the target servers.



Once complete, MemSQL Ops will show a Success page along with all the servers in the cluster.

Next, we will use MemSQL Ops can deploy the MemSQL database to the cluster but first, for non-internet connected servers, we must register the MemSQL binary to the MemSQL Ops like so:

```
sudo memsql-ops file-add -t memsql -f memsqlbin_amd64.tar.gz
```

Once complete, we can now click the DEPLOY MEMSQL button, which will bring up a screen to Confirm server's roles in the cluster. Change each role to match your individual requirements.

Deploy MemSQL > Pending MemSQL Nodes

Confirm 10 MemSQL Nodes

HIGH AVAILABILITY ADD MEMSQL NODE

MemSQL Role	Host	Port
Master Aggregator	10.0.0.165	3306
Child Aggregator	10.0.3.119	3306
Child Aggregator	10.0.1.211	3306
Leaf	10.0.0.65	3306
Leaf	10.0.0.177	3306
Leaf	10.0.3.229	3306
Leaf	10.0.0.128	3306
Leaf	10.0.3.233	3306
Leaf	10.0.2.155	3306
Leaf	10.0.2.90	3306

Shift-Click to Select Records REMOVE SELECTED EDIT SELECTED

« BACK CANCEL DEPLOY MEMSQL CLUSTER »

Click the DEPLOY MEMSQL CLUSTER button and MemSQL Ops will deploy MemSQL to the cluster in the roles designated. Once complete, you will see instructions on how to connect to your MemSQL database.

Deploy MemSQL > Deployment Successful

Deployed 10 MemSQL nodes successfully.

You can now connect to your MemSQL database by using any MySQL-compatible client. From the command line, connect using this command.

```
$ mysql -u root -h ec2-34-201-34-170.compute-1.amazonaws.com -P 3306 --prompt="memsql> ";
```

If the above command doesn't work, ensure that the host and port are correct and accessible via your firewall.

CLOSE

Congratulations! You have successfully installed and deployed MemSQL Ops and the MemSQL database!

MemSQL Ops Command Line Deployment

To deploy MemSQL Ops and the MemSQL database via the command line it is assumed one has familiarity on how to operation in a terminal environment.

To install MemSQL Ops execute the following command on the system designated as the primary:

```
tar -xzf memsql-ops-x.x.x.tar.gz
cd memsql-ops-x.x.x
sudo ./install.sh -n
```

MemSQL Ops will complete it's setup and be available on port 9000 of the designated server, i.e. http://<server IP address>:9000 but we will continue via the command line.

To avoid entering the host's IP addresses continuously for each install, it is recommended to create a file with all the IP addresses for each host like so:

```
master-agg-and-leaf-ip-10-0-0-165 /tmp $ cat memsql_hosts.txt
10.0.0.165
10.0.1.211
10.0.3.119
10.0.0.128
10.0.2.155
10.0.3.229
10.0.0.65
10.0.0.177
10.0.3.233
10.0.2.90
master-agg-and-leaf-ip-10-0-0-165 /tmp
```

To deploy MemSQL Ops we will loop through the above file to deploy to the cluster like so:

```
master-agg-and-leaf-ip-10-0-0-165 /tmp $ for i in $(cat memsql_hosts.txt)
> do
> memsql-ops agent-deploy -h $i -P 22 -u admin -i /tmp/thor
> done
There is already an agent deployed on 10.0.0.165
2017-04-19 23:34:37: J4e12c9 [INFO] Checking if host 10.0.1.211 can have an agent deployed
2017-04-19 23:34:38: J4e12c9 [INFO] We can deploy an agent to host 10.0.1.211
2017-04-19 23:34:40: J3602cb [INFO] Deploying MemSQL Ops agent to 10.0.1.211
2017-04-19 23:34:41: J3602cb [INFO] Getting MemSQL Ops binary package
2017-04-19 23:34:45: J3602cb [INFO] Copying MemSQL Ops binary to /var/lib/memsql-ops on
10.0.1.211
2017-04-19 23:34:46: J3602cb [INFO] Unpacking MemSQL Ops binary on 10.0.1.211
2017-04-19 23:34:48: J3602cb [INFO] Starting MemSQL Ops agent binary on 10.0.1.211
2017-04-19 23:34:50: J3602cb [INFO] create api client
2017-04-19 23:34:52: J3602cb [INFO] Successfully started MemSQL Ops agent binary on 10.0.1.211
```

Once MemSQL Ops is deployed successfully to the cluster, we can confirm by issuing the following command:

```
master-agg-and-leaf-ip-10-0-0-165 /tmp $ memsql-ops agent-list
 ID       Host        Port  Role      State   Version
 A875390  10.0.0.165  9000  PRIMARY   ONLINE  5.8.0
 A20d08b  10.0.3.233  9000  FOLLOWER  ONLINE  5.8.0
 A37d438  10.0.0.177  9000  FOLLOWER  ONLINE  5.8.0
 A3ab576  10.0.1.211  9000  FOLLOWER  ONLINE  5.8.0
 A77910d  10.0.2.90   9000  FOLLOWER  ONLINE  5.8.0
 A9c6276  10.0.2.155  9000  FOLLOWER  ONLINE  5.8.0
 Ab315f8  10.0.3.229  9000  FOLLOWER  ONLINE  5.8.0
 Ab3a015  10.0.0.65   9000  FOLLOWER  ONLINE  5.8.0
 Ac76692  10.0.3.119  9000  FOLLOWER  ONLINE  5.8.0
 Aea44df  10.0.0.128  9000  FOLLOWER  ONLINE  5.8.0
```

Next, we need to add our Enterprise license to the cluster to enable Enterprise features. This is accomplished with the following command:

```
master-agg-and-leaf-ip-10-0-0-165 /tmp $ memsql-ops license-add --license-key xxxxxxxxxxxxxxxxxx
Successfully added license.
```

Next, we will use MemSQL Ops can deploy the MemSQL database to the cluster but first, for non-internet connected servers, we must register the MemSQL binary to the MemSQL Ops like so:

```
master-agg-and-leaf-ip-10-0-0-165 /tmp $ sudo memsql-ops file-add -t memsql -f
memsqlbin_amd64.tar.gz
```

Now we can deploy the MemSQL database to the cluster. In this example, we will add co-located Aggregator nodes and Leaf nodes. First we add the Master Aggregator then the Child Aggregators.

Adding the Master Aggregator is done on the MemSQL Ops primary with the following command:

```
master-agg-and-leaf-ip-10-0-0-165 /tmp $ memsql-ops memsql-deploy -P 3306 -r master

2017-04-19 23:42:41: J433cf4 [INFO] Deploying MemSQL to 10.0.0.165:3306
2017-04-19 23:42:49: J433cf4 [INFO] Installing MemSQL
2017-04-19 23:42:51: J433cf4 [INFO] Finishing MemSQL Install
Waiting for MemSQL to start...
MemSQL successfully started
```

Now we can add the Child Aggregators to the cluster with the following command:

```
 for i in $(memsql-ops agent-list | grep -ivE 'primary|id'|awk '{print $1}'); do  memsql-ops
memsql-deploy -P 3306 -r aggregator -a $i; done

2017-04-19 23:46:59: J53d651 [INFO] Deploying MemSQL to 10.0.0.177:3306
2017-04-19 23:47:05: J53d651 [INFO] Downloading MemSQL from primary: 17.83%
2017-04-19 23:47:06: J53d651 [INFO] Downloading MemSQL from primary: 59.43%
2017-04-19 23:47:07: J53d651 [INFO] Downloading MemSQL from primary: 99.84%
2017-04-19 23:47:08: J53d651 [INFO] Downloading MemSQL from primary: 100.00%
2017-04-19 23:47:08: J53d651 [INFO] Installing MemSQL
2017-04-19 23:47:10: J53d651 [INFO] Finishing MemSQL Install
Waiting for MemSQL to start...
MemSQL successfully started
```

Once complete, we can move on to deploying the Leaf nodes in a similar fashion:

```
master-agg-and-leaf-ip-10-0-0-165 /tmp $ for i in $(memsql-ops agent-list -q | grep -ivE 'id' |
awk '{print $1}'); do memsql-ops memsql-deploy -a $i -P 3307 -r leaf; done

2017-04-19 23:50:41: J60acb4 [INFO] Deploying MemSQL to 10.0.0.165:3307
2017-04-19 23:50:42: J60acb4 [INFO] Installing MemSQL
2017-04-19 23:50:44: J60acb4 [INFO] Finishing MemSQL Install
Waiting for MemSQL to start...
MemSQL successfully started
```

We can confirm the successful deployment by listing all the MemSQL nodes like so:

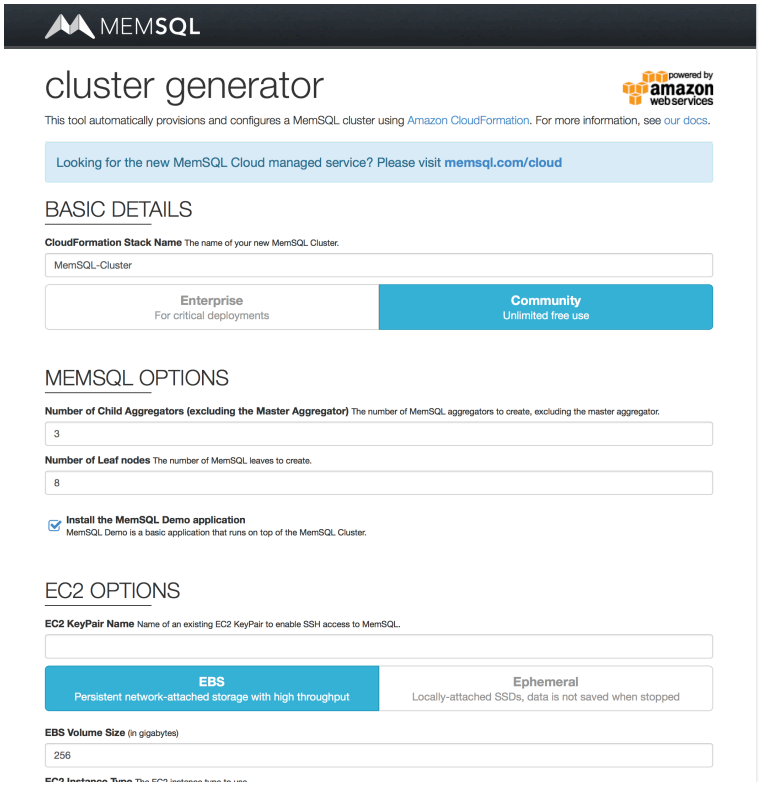
```
master-agg-and-leaf-ip-10-0-0-165 /tmp $ memsql-ops memsql-list
 ID       Agent Id  Process State  Cluster State  Role        Host        Port  Version
 8B1A7C1  A875390   RUNNING        CONNECTED      MASTER      10.0.0.165  3306  5.1.3
 0B9A1C9  A20d08b   RUNNING        CONNECTED      AGGREGATOR  10.0.3.233  3306  5.1.3
 80547B8  A37d438   RUNNING        CONNECTED      AGGREGATOR  10.0.0.177  3306  5.1.3
 E2FD9DF  A3ab576   RUNNING        CONNECTED      AGGREGATOR  10.0.1.211  3306  5.1.3
 797AE10  A77910d   RUNNING        CONNECTED      AGGREGATOR  10.0.2.90   3306  5.1.3
 A782E0B  A9c6276   RUNNING        CONNECTED      AGGREGATOR  10.0.2.155  3306  5.1.3
 EE2900E  Ab315f8   RUNNING        CONNECTED      AGGREGATOR  10.0.3.229  3306  5.1.3
 59D4067  Ab3a015   RUNNING        CONNECTED      AGGREGATOR  10.0.0.65   3306  5.1.3
 CF1C9EF  Ac76692   RUNNING        CONNECTED      AGGREGATOR  10.0.3.119  3306  5.1.3
 32072FF  Aea44df   RUNNING        CONNECTED      AGGREGATOR  10.0.0.128  3306  5.1.3
 2BB6930  A20d08b   RUNNING        CONNECTED      LEAF        10.0.3.233  3307  5.1.3
 077442B  A37d438   RUNNING        CONNECTED      LEAF        10.0.0.177  3307  5.1.3
 4FAE577  A3ab576   RUNNING        CONNECTED      LEAF        10.0.1.211  3307  5.1.3
 18E6A23  A77910d   RUNNING        CONNECTED      LEAF        10.0.2.90   3307  5.1.3
 100CA65  A875390   RUNNING        CONNECTED      LEAF        10.0.0.165  3307  5.1.3
 6457724  A9c6276   RUNNING        CONNECTED      LEAF        10.0.2.155  3307  5.1.3
 023A806  Ab315f8   RUNNING        CONNECTED      LEAF        10.0.3.229  3307  5.1.3
 C05E5FE  Ab3a015   RUNNING        CONNECTED      LEAF        10.0.0.65   3307  5.1.3
 F22756A  Ac76692   RUNNING        CONNECTED      LEAF        10.0.3.119  3307  5.1.3
 551E6D1  Aea44df   RUNNING        CONNECTED      LEAF        10.0.0.128  3307  5.1.3
```

Congratulations! You have successfully installed and deployed MemSQL Ops and the MemSQL database via the command line!

AWS CloudFormation Deployment

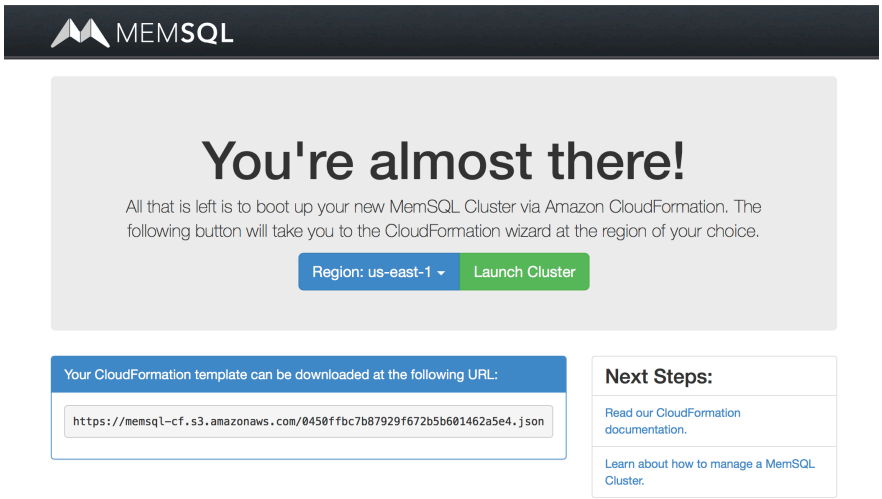
The fastest way to deploy MemSQL is on the Amazon Web Services (AWS) platform on EC2 instances using cloud.memsql.com. The only prerequisite here is that you have an existing AWS account.

1) Go to http://cloudformation.memsql.com/cloudformation
2) Click Configure and launch cluster on AWS which will bring up the MemSQL cluster generator, which will bring up a webpage similar to the screenshot below:



3) Fill in the necessary fields on the form, which include:
    a. MemSQL license key
        i. Community Edition users are not required to provide a license key
    b. Your Amazon EC2 KeyPair name
        i. See the Amazon documentation for generating your own keypair.
        ii. The EC2 key pair must be associated with the availability zone where you plan to launch your cluster.
4) Click the Generate Template button

5) In the following screen, look for the CloudFormation template generated. You should see a URL that leads to a downloadable JSON file.



6) Choose your desired Amazon EC2 availability zone and select Launch Cluster. This will take you to the AWS website where you can use the CloudFormation template to fully launch your cluster.
7) After the cluster is launched, you will be redirected to the CloudFormation summary page. Click the refresh button in the top right corner. You can use this page to monitor your cluster.
8) Once your cluster is up and running, SSH to the master aggregator using the user "ubuntu" and the keypair you specified in the config. From the master aggregator you can use the memsql command line tool to open a MySQL client.

```
$ssh -i <path to EC2 private key> ubuntu@<aggregator hostname>
$memsql
```

9) Once logged in with the MySQL client, you can run DDL and DML commands.
10) Verify that connecting to http://<hostname master agg>:9000 brings you to the latest Ops. If it does not, ensure that your AWS settings allow you to expose port 9000.

You can now start creating databases and querying MemSQL!

MEMSQL

Enabling High Availability

In MemSQL, the availability layer is responsible for mapping partitions to leaves (both masters and slaves) and is managed through a series of SQL commands that let you inspect and modify its state.

MemSQL exposes two modes of operation: redundancy-1 and redundancy-2.
Redundancy-1:

- There are no extra online copies of your data (all partitions are masters). In the event of a leaf node failure, your data is offline until you reintroduce the leaf back into the system. If your leaf node is irrecoverably lost, you can use the REBALANCE PARTITIONS … FORCE command to create empty replacement partitions instead of recovering your data.

Redundancy-2:

- MemSQL handles node failures by promoting the appropriate slave partitions into masters so that your databases remain online. Of course, if all of your machines fail, then your data is unavailable until you recover enough machines or recreate the cluster from scratch. Each redundancy mode has an expected, balanced state.

To provide the highest levels of availability in a MemSQL cluster, it is highly recommended to enable High Availability. During this process, Ops will move half of the leaves in the cluster from the availability group 1 into the availability group 2. This is considered an offline operation. As all partitions of all databases will be duplicated, each leaf node needs at least 50% free memory and disk space. In addition, the cluster must have an even number of leaves.

If you need to add more nodes to the cluster, we recommend doing it before enabling High Availability.

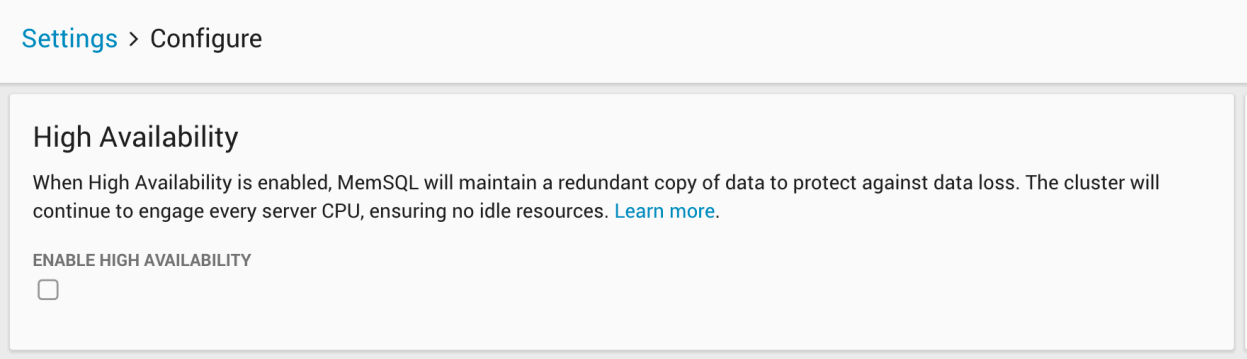
Note that currently MemSQL Ops does not support disabling High Availability.

MemSQL Ops GUI

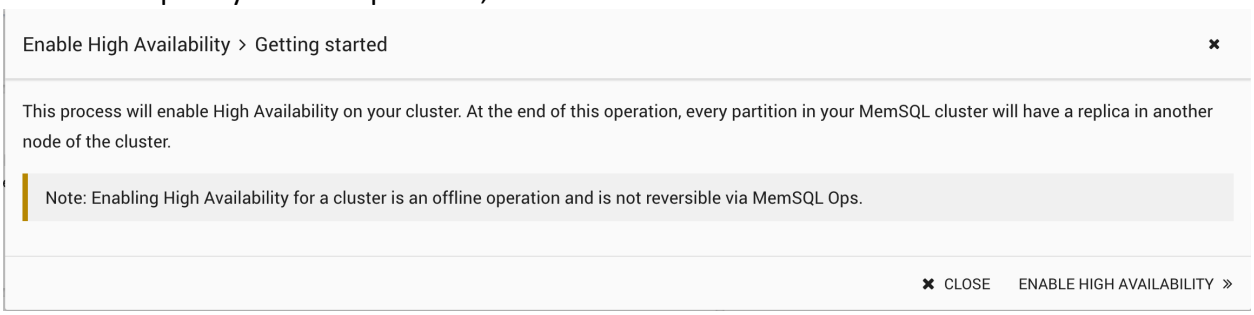
The following steps will enable High Availability for in the cluster.

In your browser, enter the following into the address bar:
http://<server IP address>:9000/settings/config

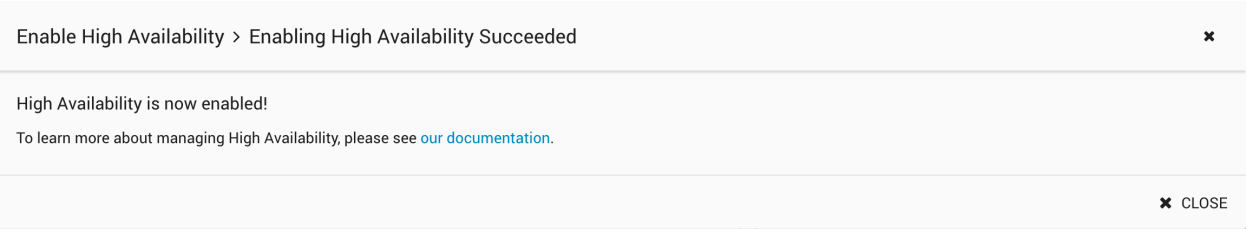
Once in the Configure screen for the cluster you will see the sections for enabling High Availability.



Click the check box to begin the process for setting up High Availability. When prompted that this is a temporary off-line operation, click the 'ENABLE HIGH AVAILABITLY' button.



MemSQL Ops will enable High Availability and you will see the following screen confirming success.



Congratulations! You are now running in High Availability mode!

MemSQL Ops Command Line

The following command enables High Availability for in the cluster:

```
memsql-ops memsql-enable-high-availability --no-prompt

Enabling high availability
Currently on state EnableHighAvailability.EnableHighAvailabilityInit
Currently on state EnableHighAvailability.UpdatingConfigs
Currently on state EnableHighAvailability.UpdatingLeafGroups
Currently on state EnableHighAvailability.RestoringRedundancy
Currently on state EnableHighAvailability.EnableHighAvailabilitySuccess
Successfully enabled high availability.
```

Congratulations! You are now running in High Availability mode!

MEMSQL

Running the Sample Data Set

To explore running SQL commands in MemSQL we have packaged an example data set for use. The data set can be downloaded from:
https://s3.amazonaws.com/mdemarco-s3/sgpoc/nyc_taxi.tar
or contact your local Sales Engineer to provide you a copy.

Uncompressing the data set

Copy the file to the MemSQL Master Aggregator and uncompress the file with the following command:

```
tar xvf nyc_taxi.tar
nyc_taxi/
nyc_taxi/ReadMe.txt
nyc_taxi/nyc_taxi_data.sql
nyc_taxi/nyc_taxi_queries.sql
nyc_taxi/nyc_taxi_schema.sql
```

Once the file is uncompressed, you will have the new directory name nyc_taxi with the following files:

`ReadMe.txt` – file describing data set files and table counts
`nyc_taxi_schema.sql` – database/table creation script
`nyc_taxi_data.sql` – data for loading into nyc_taxi schema
`nyc_taxi_queries.sql` – example queries

Creating the database schema and tables

To create the database schema and tables for the data set issue the following command:

```
cd nyc_taxi; memsql <nyc_taxi_schema.sql
```

Loading data

To load data into the database issue the following command:

```
memsql <nyc_taxi_data.sql
```

To confirm you have the data loaded correctly, issue the following command:

```
memsql nyc_taxi -e'select count(*) Count_of_Trips from trips;select
count(*) Count_of_Drivers from drivers;select count(*)
Count_of_Neighborhoods from neighborhoods;'
```

This will yield the following table counts:

```
+----------------+
| Count_of_Trips |
+----------------+
|          90389 |
+----------------+
+------------------+
| Count_of_Drivers |
+------------------+
|             2500 |
+------------------+
+------------------------+
| Count_of_Neighborhoods |
+------------------------+
|                    612 |
+------------------------+
```

Querying the data set

To query the data set issue the following command:

```
memsql <nyc_taxi_queries.sql
```

To examine the queries, simply `more` the file like so:

```
more nyc_taxi_queries.sql
```

To run a more interactive query session, you can log into the MemSQL database like so:

```
 memsql nyc_taxi
```

You can run the query file with the following command:

```
memsql> source nyc_taxi_queries.sql
```

You can list the tables in the database with the following command:

```
memsql> show tables;
+--------------------+
| Tables_in_nyc_taxi |
+--------------------+
| drivers            |
| neighborhoods      |
| trips              |
+--------------------+
3 rows in set (0.00 sec)
```

You can describe the structure of any of the tables using the `describe` command like so:

```
memsql> describe drivers;
+---------------+----------------+------+------+---------+-------+
| Field         | Type           | Null | Key  | Default | Extra |
+---------------+----------------+------+------+---------+-------+
| id            | bigint(20)     | NO   | MUL  | NULL    |       |
| first_name    | varchar(128)   | YES  |      | NULL    |       |
| last_name     | varchar(128)   | YES  |      | NULL    |       |
| location      | geographypoint | NO   | MUL  | NULL    |       |
| goal_location | geographypoint | YES  |      | NULL    |       |
| status        | varchar(20)    | YES  |      | NULL    |       |
| trip_id       | bigint(20)     | YES  |      | -1      |       |
+---------------+----------------+------+------+---------+-------+
7 rows in set (0.00 sec)
```

When finished exploring the database, you can exit your session with the `exit` command like so:

```
memsql> exit
Bye
```

Congratulations! You have successfully explored a data set in MemSQL!

Commands Used

Command for deploying MemSQL Ops agent to additional hosts:

```
for i in $(cat memsql_hosts.txt)
do
memsql-ops agent-deploy -h $i -P 22 -u admin -i /tmp/thor
done
```

Command for listing MemSQL Ops agents:

```
memsql-ops agent-list
```

Command for adding MemSQL Enterprise license file:

```
memsql-ops license-add --license-key xxxxxxxxxxxxxxxxx
```

Command for deploying MemSQL Master Aggregator:

```
memsql-ops memsql-deploy -P 3306 -r master
```

Command for deploying MemSQL Child Aggregators:

```
for i in $(memsql-ops agent-list | grep -ivE 'primary|id'|awk '{print $1}')
do
memsql-ops memsql-deploy -P 3306 -r aggregator -a $i
done
```

Command for deploying MemSQL Leaf Nodes:

```
for i in $(memsql-ops agent-list -q | grep -ivE 'id' | awk '{print $1}')
do
memsql-ops memsql-deploy -a $i -P 3307 -r leaf
done
```

Command for listing MemSQL status in cluster:

```
memsql-ops memsql-list
```

Command for enabling High Availability:

```
memsql-ops memsql-enable-high-availability --no-prompt
```

Command for uncompressing file:

```
tar xvf nyc_taxi.tar
```

Command for creating demo schema:

```
memsql <nyc_taxi_schema.sql
```

Command for loading data into demo schema:

```
memsql <nyc_taxi_data.sql
```

Command for checking data loads:

```
memsql nyc_taxi -e'select count(*) Count_of_Trips from trips;select count(*) Count_of_Drivers from drivers;select count(*) Count_of_Neighborhoods from neighborhoods;'
```

Command for running demo queries:

```
memsql <nyc_taxi_queries.sql
```

Command for examining query file:

```
more nyc_taxi_queries.sql
```

Command for logging into MemSQL demo schema:

```
memsql nyc_taxi
```

Command for running query file from within MemSQL:

```
source nyc_taxi_queries.sql;
```

Command to show tables:

```
show tables;
```

Command to describe table structure:

```
describe drivers;
```

Command to end MemSQL session:

```
exit;
```

MEMSQL

Document Control

Title:	Self-Guided Proof of Concept Documentation
Author	Matt DeMarco (matt@memsql.com)
Control	Reissue as complete document only

Document Change Record

Date	Version	Author	Change Details
07.11.2017	Version 1	Matt DeMarco	Initial Version
07.12.2017	Version 2	Matt DeMarco	Added MemSQL beta 6 install instructions and feature list
07.12.2017	Version 3	Matt DeMarco	Added self-guided PoC instructions. Added command list in appendix. Added High Availability section
07.14.2017	Version 4	Matt DeMarco	Correct formatting errors
08.28.2017	Version 5	Matt DeMarco	Changed Ops install procedure as to not install MemSQL
11.3.2017	Version 6	Matt DeMarco	Updated for MemSQL 6
05.24.2018	Version 7	Matt DeMarco	Updated language